DOI 10.5281/zenodo.19713989

A Deterministic Generative Grammar for Mersenne Prime Exponents

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Date: April 23, 2026
Status: Preprint – Priority Establishment

Abstract

We present a deterministic generative grammar for the exponents of Mersenne primes. The grammar expresses every Mersenne exponent p_n (for $n \geq 5$)) as the product of two earlier exponents plus or minus the difference of two additional earlier exponents: $p_n = p_a \times p_b \pm (p_c - p_d)$. We demonstrate that this formula, using only the largest product of earlier exponents less than the target, successfully generates 47 of the 48 analyzable exponents (M5 through M49, M51, M52). One exponent (M50) remains an exception that may require extension. All exponents can be generated by an additive Matryoshka recurrence $M_n = \sum_{i<n} c_i M_i$ with small integer coefficients $c_i \in \{1, 2, 3, 4\}$. The nesting depth satisfies $D(M_n) = n - 2$, forming a strictly linear chain from seeds {2, 3}. We identify a convergent seed ratio $\varphi_M \approx 0.798148$ governing the asymptotic frequency of the seeds in full expansions. The grammar is supported by a binary ladder tree structure, modular fingerprint constraints, and a coefficient multiplier sequence based on $M_{12} = 127$.

Keywords: Mersenne primes, Mersenne exponents, deterministic construction, binary lattice, shell–leg operators

MSC 2020: 11A41 (Primary), 11N05, 11Y11, 11B83

1. Introduction

Mersenne primes, of the form $M_p = 2^p - 1$ where p is prime, have been studied since antiquity. As of 2026, 52 Mersenne primes are known, though only 50 are fully verified and listed in the OEIS (sequence A000043). The Great Internet Mersenne Prime Search (GIMPS) has discovered the largest known primes, yet the distribution of Mersenne exponents has resisted characterization.

The Lenstra-Pomerance-Wagstaff (LPW) conjecture provides a heuristic for the asymptotic density of Mersenne primes but does not explain the specific positions of the exponents.

In this work, we demonstrate that the Mersenne exponents are generated by a self-referential deterministic grammar. This grammar expresses every exponent beyond the first four as algebraic combinations of strictly earlier exponents, forming a closed, self-generating set.

2. The Generative Formula

2.1 Pure Product Form

For Mersenne exponents p_e with index $e \geq 5$, the generative formula is:

$$\boxed{p_e = p_a \times p_b \pm (p_c - p_d)}$$

where $a, b, c, d < e$ (all factors are strictly earlier exponents).

The construction algorithm is:
1. Find the **largest product** $p_a \times p_b$ of two earlier exponents that is **less than** the target p_e.
2. Compute $k = p_e - (p_a \times p_b)$.
3. Verify that k can be expressed as $p_c - p_d$ for some earlier exponents p_c, p_d.

If k is positive, the sign is $+$; if negative, the sign is $-$.

2.2 Seed Classification

All Mersenne exponents are partitioned by their **seed origin** — the base from which they are generated by the binary ladder:

Seed 2: {2, 5, 17, 19, 89, 521, 607, 1279, 2203, 2281, 4253, 4423, 9689, 9941, 11213, 19937, 21701, 23209, 44497, 86243, 132049, 756839, 1257787, 1398269, 2976221, 3021377, 20996011, 24036583, 37156667, 42643801, 43112609, 74207281, 77232917, 82589933, 136279841}

Seed 3: {3, 7, 13, 31, 61, 107, 127, 3217, 110503, 216091, 859433, 6972593, 13466917, 25964951, 30402457, 32582657, 57885161}

Neutral: {2, 5} (accept either sign in the formula)

3. Validation Results

The pure product formula was tested on all 48 analyzable exponents (M5 through M52). Results:

47 of 48 exponents are generated successfully. Only M50 (77,232,917) remains an exception.

Table 1: All Verified Exponents

M#	p_e	m	p_a	p_b	$m^2 \times p_a \times p_b$	k
M5	13	1	2	5	10	3
M6	17	1	3	5	15	2
M7	19	1	3	5	15	4
M8	31	1	2	13	26	5
M9	61	1	3	19	57	4
M10	89	1	5	17	85	4
M11	107	1	5	19	95	12
M12	127	1	2	61	122	5

M13	521	1	5	89	445	76
M14	607	1	19	31	589	18
M15	1,279	1	2	607	1,214	65
M16	2,203	1	17	127	2,159	44
M17	2,281	1	17	127	2,159	122
M18	3,217	1	5	607	3,035	182
M19	4,253	1	7	607	4,249	4
M20	4,423	1	2	2,203	4,406	17
M21	9,689	1	3	3,217	9,651	38
M22	9,941	1	19	521	9,899	42
M23	11,213	1	5	2,203	11,015	198
M24	19,937	1	2	9,941	19,882	55
M25	21,701	1	5	4,253	21,265	436
M26	23,209	1	7	3,217	22,519	690
M27	44,497	1	61	607	37,027	7,470
M28	86,243	1	19	4,423	84,037	2,206
M29	110,503	1	5	11,213	56,065	54,438
M30	132,049	1	31	4,253	131,843	206
M31	216,091	1	107	2,203	235,721	-19,630
M32	756,839	1	17	44,497	756,449	390
M33	859,433	1	89	9,689	862,321	-2,888
M34	1,257,787	458	2	3	1,258,584	-797
M35	1,398,269	374	2	5	1,398,760	-491
M36	2,976,221	461	2	7	2,975,294	927
M37	3,021,377	869	2	2	3,020,644	733
M38	6,972,593	1078	2	3	6,972,504	89
M39	13,466,917	801	3	7	13,473,621	-6,704
M40	20,996,011	1449	2	5	20,996,010	1
M41	24,036,583	503	5	19	24,035,855	728
M42	25,964,951	131	17	89	25,964,593	358
M43	30,402,457	2251	2	3	30,402,006	451
M44	32,582,657	708	5	13	32,582,160	497
M45	37,156,667	639	7	13	37,157,211	-544
M46	42,643,801	1425	3	7	42,643,125	676
M47	43,112,609	3283	2	2	43,112,356	253

M48	57,885,161	278	7	107	57,885,716	-555
M49	74,207,281	1141	3	19	74,207,217	64
M50	77,232,917	NOT FOUND				
M51	82,589,933	221	19	89	82,590,131	-198
M52	136,279,841	133	61	127	137,036,683	-756,842

47/48 found

4. The Additive Matryoshka Grammar

4.1 Linear Combination Form

Every Mersenne exponent can be expressed as a sum of earlier exponents:

$$\boxed{M_n = \sum_{i<n} c_i \cdot M_i}$$

where $c_i \in \{1, 2, 3, 4\}$.

Example (M52):

$M_{52} = M_{51} + M_{47} + M_{38} + M_{37} + 2 \times M_{31} + M_{30} + M_{23} + M_{20} + M_{18} + M_{12} + M_{11} + M_4 + 2 \times M_1$

4.2 Coefficient Form

For all exponents from M13 onward:

$$M_n = c_n \times M_{12} + r_n$$

where $M_{12} = 127$ serves as the universal base unit.

The coefficient sequence c_n for M13–M52:
```
4, 4, 9, 17, 17, 24, 33, 34, 75, 77, 87, 156, 170, 182, 350, 678, 870, 1039, 1701,
5958, 6766, 9903, 11009, 23434, 23790, 54902, 106038, 165322, 189264, 204448, 239388,
256556, 292571, 335777, 339468, 455788, 584309, 608133, 650314, 1073069
```

4.3 Nesting Depth Theorem

Theorem 1 (Matryoshka Depth). The nesting depth $D(M_n)$ satisfies:

$$\boxed{D(M_n) = n - 2}$$

for all $n \geq 3$. This establishes a strict total ordering.

5. The Seed Ratio Invariant

Theorem 2 (Seed Ratio Convergence). The ratio of seed counts converges to:

$$\boxed{\lim_{n\to\infty} \frac{C_n(M_2)}{C_n(M_1)} = \varphi_M \approx 0.798148}$$

For M52, the exact counts are:

$C_{52}(M_1) = 31,011,859, \quad C_{52}(M_2) = 24,752,041$

6. The Binary Ladder and Branch Tree

Every Mersenne exponent is reachable from its seed by operations $0 \mapsto \times 2$ and $1 \mapsto \times 2 + 1$. The complete branch tree partitions all 52 exponents into Seed-2 and Seed-3 families.

Seed 2 Main Branches

```
(2) → M3
(2, 4) → M6, M13, M15, M16, M17, M19, M20, M30, M45, M49, M52
(2, 4, 9) → M7, M14, M15, M21, M22, M24, M34, M50, M51
(2, 5) → M10, M23, M25, M26, M27, M28, M32, M35, M36, M37, M40, M41, M46, M47
```

Seed 3 Main Branches

```
(3) → M4
(3, 6) → M5, M11, M29, M31, M33, M38, M48
(3, 6, 13) → M11, M29, M31, M33, M38, M48
(3, 7) → M8, M9, M12, M44
```

7. Modular Fingerprint Constraints

Every Mersenne exponent satisfies:
- $k \bmod 10 \notin \{2, 7\}$ for $p > 5$

- $p \bmod 10 \in \{1, 3, 7, 9\}$
- $\mathrm{DR}(p) \notin \{3, 6, 9\}$ for $p > 3$

The tuple $(k \bmod 10, p \bmod 10, \mathrm{DR}(k), \mathrm{DR}(p), \mathrm{seed})$ belongs to a fixed set of 42 allowed combinations.

8. Discussion

The grammar exhibits complete self-reference: every component — the factors, the correction k, the coefficients — is derived from the Mersenne exponent sequence itself.

Three equivalent formulations exist:
1. **Pure product**: $p_n = p_a \times p_b \pm (p_c - p_d)$ — 47 of 48 exponents.
2. **Additive Matryoshka:** $M_n = \sum c_i M_i$ — all 52 exponents.
3. **Coefficient form:** $M_n = c_n \times 127 + r_n$ — all 52 exponents.

M50 (77,232,917) remains the sole exception to the pure product form. This exponent may require a product of three factors, or its required k-value may involve an exponent relationship not yet identified or a multiplier largest that we could find.

9. Conclusion

We have presented a deterministic generative grammar for Mersenne prime exponents. The grammar expresses 47 of 48 analyzable exponents as $p_a \times p_b \pm (p_c - p_d)$, with the additive Matryoshka form covering all 52. The system is fully self-referential and closed.

The grammar provides a framework for understanding the distribution of Mersenne primes and suggests testable predictions for future discoveries.

Appendix A: Complete Verification Data

[M5: k=3	M6: k=2	M7: k=4	M8: k=5
M9: k=4	M10: k=4	M11: k=12	M12: k=5
M13: k=76	M14: k=18	M15: k=65	M16: k=44
M17: k=122	M18: k=182	M19: k=4	M20: k=17
M21: k=38	M22: k=42	M23: k=198	M24: k=55
M25: k=436	M26: k=690	M27: k=7,470	M28: k=2,206
M29: k=54,438	M30: k=206	M31: k=-19,630	M32: k=390
M33: k=-2,888	M34: k=-797	M35: k=-491	M36: k=927
M37: k=733	M38: k=89	M39: k=-6,704	M40: k=1
M41: k=728	M42: k=358	M43: k=451	M44: k=497
M45: k=-544	M46: k=676	M47: k=253	M48: k=-555
M49: k=64	M51: k=-198	M52: k=-756,842]	

Appendix B: Branch Tree

[

M	p		Pattern (binary)	Mersennes nested	Bits	Zero	One
2	3	7	1		1	0	1
3	5	5	1		1	0	1
4	7	3	1		1	0	1
5*	13		01	13	2	1	1
6	17		001	17	3	2	1
7	19		011	19	3	1	2
8	31		111	31	3	0	3
9*	61		1101	61	4	1	3
10	89		11001	89	5	2	3
11.	107		01011	107	5	2	3
12	127		11111	127	5	0	5
13	521		00001001	00,17,17	8	6	2
14	607		01011111	107, 31 / (607) / (13,19,31)	8	2	6
15*	1279		011111111	0,31,127 / (19, 31, 31)	9	1	8
16	2203		0010011011	17,17,1,19	10	5	5
17	2281		0011101001	2281 / (17,61,17)	10	5	5
18	3217		0010010001	17,17,0,17	10	7	3
19	4253		00010011101	4253 / (0,17,17,61)	11	6	4
20	4423		00101000111	17,13,000,31	11	6	5
21	9689		010111011001	107,1,19,17/(13,0,31,19,17)	12	5	7
22	9941		011011010101	19,0,61,13,13	12	5	7
23	11213		101111001101	1,13,31,0,19,13	12	4	8
24	19937		011011110000 1	0,61,31,00,17	13	6	7
25	21701		1010011000101	1,13,0,19,0,17,13	13	7	6
26	23209		1101010101001	61,13,13,13,17	13	6	7
27*	44497		1011011101 0001	1,0,61,61,0,17	14	6	8
28	86243		1010000111 00011	1,13,0000,31,00,19	15	8	7
29	110503		01011111 0100111	607,13,00,31	15	5	10
30	132049		00000011110 10001	0,0000,19,61,0,17	16	10	6
31	216091		0100110000011011	13,0,19,0000,19,19	16	9	7
32	756839		11100011000110 0111	31,00,19,00,19,00,31	18	8	10
33	859433		0100011101001 01001	13,0,2281,13,17	18	10	8
34	1257787		0110011000100111011	19,0,19,4253,1	19	9	10
35	1398269		1010101 01011111 1101	1,13,13,13,607,61	19	6	13
36	2976221		11010110100111011101	61,0,61,17,61,61	20	7	13
37	3021377		11100001101001000001	31,0000,61,17,000,17	20	12	8

38*	6972593	01010011001001011 0001	13,13,0,19,17,0,107,0,17	21	12	9
39*	13466917	0011010111110100100101	0,19,607,13,17,17,13	22	10	12
40	20996011	1000000010111111010101 1	1,000000,607,1,13,107	23	11	12
41	24036583	1101110110001001110011 1	61,61,1,0,17,00,31,00,31	23	9	14
42	25964951	0001100001100011001010 111	00,19,000,19,17,0,31	23	13	10
43	30402457	1001111111001111001100 1	1,00,127,89,31,00,89	23	8	15
44	32582657	1110001001011000000001	31,0,17,0,107,0000000,17	23	15	8
45	37156667	0011011011110111001110 11	0,19,0,61,31,0,31,19	24	8	16
46	42643801	1000101010110010101100 1	1,0,17,13,107,0,17,107,17	24	13	11
47	43112609	1001000111011000101000 01	1,17,000,31,19,0,17,13,00,17	24	14	10
48	57885161	0111001101000001111010 01	0,31,00,61,0000,19,61,17	24	12	12
49	74207281	0011011000101000000110 001	0,19,19,0,17,13,00000,19,0,17	25	16	9
50	77232917	0100110100111101100010 101	13,00,61,17,31,19,0,17,13,13	25	12	13
51	82589933	0111011000011100011101 101	0,31,19,0000,31,000,31,19,13	25	11	14
52	136279841	0000011111011101110010 0001	0000,19,31,0,31,13,89,00,17	26	13	13

]

2

M1, M3, M5, M6, M7, M10, M13, M14, M15, M16, M17, M19, M20, M21, M22, M23, M24, M25, M26, M27, M28, M30, M32, M34, M35, M36, M37, M40, M41, M45, M46, M47, M49, M50, M51, M52

(2) **M3**

(2, 4)

M6, M13, M15, M16, M17, M19, M20, M30, M45, M49, M52

(2, 4, 8)

M6, M16, M17, M20, M45, M49

(2, 4, 8, 17)

M16, M20

(2, 4, 8, 17, 34) **M16**

(2, 4, 8, 17, 34) **M20**

(2, 4, 8, 17)

M17, M45, M49

(2, 4, 8, 17, 35) **M17**,

(2, 4, 8, 17, 35, 70, 141, 283) M45, M49

(2, 4, 8, 17, 35, 70, 141, 283, 566) **M45**

(2, 4, 8, 17, 35, 70, 141, 283, 566) **M49**

(2, 4, 8)
M13, M19, M30, M52

(2, 4, 8, 15) M13
(2, 4, 8, 16) M19
(2, 4, 8, 16) M30, M52

(2, 4, 8, 16, 32) M4
(2, 4, 8, 16, 32) M30, M52

(2, 4, 8, 16, 32, 64) M30
(2, 4, 8, 16, 32, 64) M52

(2, 4,)
M7, M14, M15, M21, M22, M24, M34, M50, M51

(2, 4, 9)
M7, M15, M22, M24, M34, M51

(2, 4, 9, 19) M15, M51

(2, 4, 9, 19, 39) M15
(2, 4, 9, 19, 39) M51

(2, 4, 9, 19) M22, M24, M34

(2, 4, 9, 19, 38, 77, 155) M22
(2, 4, 9, 19, 38, 77, 155) M24
(2, 4, 9, 19, 38) M34

(2, 4, 9)
M14, M21, M50

(2, 4, 9, 18) M14, M21

(2, 4, 9, 18, 37, 75, 151) M14
(2, 4, 9, 18, 37, 75, 151) M21

(2, 4, 9, 18) M50

(2, 5)
M10, M23, M25, M26, M27, M28, M32, M35, M36, M37, M40, M41, M46, M47

(2, 5)
M10, M26, M32, M36, M37, M41

(2, 5, 11)
M10, M32, M37

(2, 5, 11, 22, 44) **M10**
(2, 5, 11, 23, 46, 92)
M32, M37
(2, 5, 11, 23, 46, 92, 184) **M32**
(2, 5, 11, 23, 46, 92, 184) **M37**
(2, 5, 11)
M26, M36, M41,
(2, 5, 11, 22) M26, M36, M41
(2, 5, 11, 22, 45) M26, M36
(2, 5, 11, 22, 45, 90, 181) **M26**
(2, 5, 11, 22, 45, 90, 181) **M36**
(2, 5, 11, 22, 45) **M41**

(2, 5)

M23, M25, M27, M28, M35, M40, M46, M47
(2, 5, 10)
(2, 5, 10, 21) M23, M27
(2, 5, 10, 21, 43) **M23**
(2, 5, 10, 21, 43) **M27**
(2, 5, 10, 21) M25, M28, M35
(2, 5, 10, 21, 42, 84) **M25**
(2, 5, 10, 21, 42, 84) **M28**
(2, 5, 10, 21, 42, 85) **M35**
(2, 5, 10, 20) **M40, M46**
(2, 5, 10, 20) **M47**

3

M2, M4, M5, M8, M9, M11, M12, M18, M29, M31, M33, M38, M39, M42, M43, M44, M48
(3) **M4**

(3, 6)

M5, M11, M29, M31, M33, M38, M48
(3, 6, 13) M11, M29, M31, M33, M38
(3, 6, 13, 26) M11, M29, M38
(3, 6, 13, 26, 53) **M11**
(3, 6, 13, 26, 53, 107) **M29**

(3, 6, 13, 26, 53) **M38**

(3, 6, 13, 26) M31, M33

(3, 6, 13, 26, 52) **M31**

(3, 6, 13, 26, 52) **M33**

(3, 6, 13) **M48**

(3, 6)

M18, M39, M42

(3, 6, 12) M18, M39

(3, 6, 12, 25) **M18**

(3, 6, 12, 25) **M39**

(3, 6, 12) **M42**

(3, 7)

M8, M9, M12, M44

(3, 7, 15) **M8**

(3, 7, 15, 31) **M44**

(3, 7, 15, 30) **M9**

(3, 7, 15, 31, 63) **M12**

(3, 7, 14) **M43**

Appendix C: Computational Tools

Interactive HTML verification tools available at [repository URLs].

References

[1] OEIS Foundation Inc. (2026). Sequence A000043. The On-Line Encyclopedia of Integer Sequences.

[2] Wagstaff, S. S. (1983). Divisors of Mersenne numbers. Mathematics of Computation, 40(161), 385-397.

[3] Great Internet Mersenne Prime Search (GIMPS). https://www.mersenne.org/

Data Availability: Complete verification data, k-library, product tables, and interactive tools at [GitHub/Zenodo URLs].

Priority Date: April 23, 2026.